

Programming Fundamentals

Lab 12

Question # 1

10 Marks

Write a C program which prints the following window pattern after taking *height* and *width* from the user. For example, if the user enters height to be 5 and width to be 11, the following pattern should be displayed on screen:

```
+++++  
+ + + + +  
+++++  
+ + + + +  
+++++
```

In order to display the above pattern, your program should use ONLY following THREE printf() statements:

```
printf("+");    printf(" ");    printf("\n");
```

Input validation: Your program should make sure that both height and width are **odd numbers**. Moreover, the value of **height should be at least 3** and value of **width should be at least 5**. In case of invalid input, your program should keep prompting the user again and again till the user provides valid input.

Question # 2

10 Marks

Write a C program which prints the following square pattern by taking the size (dimensions) from the user. For example, if the user enters the size to be 6, the following 6 by 6 patterns should be displayed on screen:

```
111111  
011111  
001111  
000111  
000011  
000001
```

In order to display the above pattern, your program should use ONLY following THREE printf() statements:

```
printf("+");    printf(" ");    printf("\n");
```

Input validation: Your program should make sure that the size is an **even number greater than or equal to 2**. In case of invalid input, your program should keep prompting the user again and again till the user provides valid input.

Question # 3

10 Marks

Define a C function having three parameters. This function should take an array of integers and its size as its first two parameters. This function should determine the most frequent element of the array, and it should return the **most frequent element** (i.e. the element which occurs the most often) and its **frequency**, should be displayed in the function. In your function, you are NOT allowed to declare or use any other array. Illustrate the working of your function (on various inputs) in a complete C program. See the following examples:

Example #	Elements of the array	Most frequent element	Frequency
1	5 3 4 5 2 7 5 4	5	3
2	12 74 23 35 23 12	12 or 23	2
3	7 5 6	7 or 5 or 6	1

As you can see from the last two examples, if more than one element is the most frequent element, then your function can determine any one of them as the most frequent element.

Illustrate the working of your function (on at least 5 different examples/inputs) in a complete C program.

Question # 4

10 Marks

Define a C function which should remove all NEGATIVE values from an unsorted array of integers. Your function should also preserve the order of remaining elements in the array. At the end, this function should also return the count of the NEGATIVE values that were removed from the array. The prototype of your function should be:

int removeNegatives (int arr[], int size)

For example, if **arr** contains these 7 elements { 11, -15, -2, 7, 11, 6, -8 }, then, after the function call to **removeNegatives** the **arr** should now contain { 11, 7, 11, 6 } in its first four indices. The function should return 3 (because, 3 negative values were removed from the array).

Important Note:

- In your function, you are NOT allowed to declare or use any other array.

Illustrate the working of your function (on at least 4 different examples/inputs) in a complete C program. New Size will be originalSize-countOfNegatives you should display that too like previous lab Task of removing duplicates.

Question # 5

10 Marks

Define a C function:

void cyclicRotate (int arr [], int n, int k);

This function takes an array (**arr**) and its size (**n**) as its first two parameters. Third parameter is an integer value **k**. This function should *cyclically rotate* all the elements of the array **arr**, by **k** positions to the right.

For example, if **arr** contains these **8** elements { **3, 4, 5, 8, 7, 2, 9, 1** }, then, after the function call **cyclicRotate (arr, 8, 3)** the **arr** should contain these **8** elements { **2, 9, 1, 3, 4, 5, 8, 7** }.

In your function:

- You can assume that **k** is greater than **0** and less than **n**
- You are NOT allowed to declare or use any other array.

Illustrate the working of your function (on at least 4 different examples/inputs) in a complete C program.

Hint 1: What happens if **k** is **1**?

Hint 2: You may implement another helper function to accomplish this task.

Hint 3: Do some paperwork.

These *good programming practices* will also have their (significant) weightage in the marking of this assignment:

- Comment your code intelligently. **Uncommented code will NOT be given any credit.**
- Indent your code properly.
- Use meaningful variable and function names. Use **camelCase** notation.
- Use meaningful prompt lines/labels for input/output.

😊 **GOOD LUCK!** 😊